

Multiple Choice (10 marks)

1. Statement 1— The Stanford Sentiment Treebank contained movie reviews, not book reviews. Statement 2— The Penn Treebank has been used for language modeling.
 - (a) True, True
 - (b) False, True
 - (c) True, False
 - (d) True, Not Given
 - (e) False, Not Given
2. Let $x = 1$. What is $x \uparrow\uparrow 3$ in Python 3?
 - (a) 16
 - (b) 4
 - (c) 8
 - (d) 5
 - (e) 10
3. Consider $x(t)$ to be given as, $x(t) = 10 \cos(20\pi t) - 5 \cos(50\pi t)$ What is minimum sampling rate (/Hz) such that $y(t) = x(t)$?
 - (a) 30
 - (b) 20
 - (c) 40
 - (d) 50
 - (e) 35
4. Which types of functions grow the slowest?
 - (a) $O(N)$
 - (b) $O(\log \log N)$
 - (c) $O(N!)$
 - (d) $O(\log N)$
 - (e) $O(N^N)$
5. Given a color image of size $28 \times 28 \times 3$ pixels, how many convolutional filters in the first layer of a Convolutional Neural Network if the first layer's...
 - (a) 96

- (b) 512
- (c) 1024
- (d) 64
- (e) 48

True / False (10 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Statement 1— The back-propagation algorithm learns a globally optimal neural network with hidden layers. Statement 2— The VC dimension of...' is 'True, True'.
2. True or False: The correct answer to 'Let $l = [1,2,3,4]$. What is `sum(l)` in Python3?' is '10'.
3. True or False: The correct answer to 'Obtain the 15's and 16's complement of the following hexadecimal numbers: a) FFOb) 1234c) ABCDd) 0000' is 'a) FFF, b) 1234, c) ABCD, d) 0000'.
4. True or False: The correct answer to 'Of the following, which gives the best upper bound for the value of $f(N)$ where f is a solution to the recurrence $f(2N + 1) = f(2N) = f(N)...$ ' is ' $O((\log N)^2)$ '.
5. True or False: The correct answer to 'Calculate the Hamming pairwise distances and determine the minimum Hamming distance among the following codewords:...' is '1'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. In Chord, assume the size of the identifier space is 16. The active nodes are N3, N6, N8 and N12. Show all the target key (in ascending order, ignore the node's identifier itself) for N6.
2. A computer program uses 3 bits to represent integers. When the program adds the decimal (base 10) numbers 5 and 3, the result is 0. Which of the following is the best explanation for the result?

End of Paper